

The integration of security practices into SDLC for banking web applications in South Africa.

Secure software development in banking applications.

Introduction/background

Digital technology has transformed rapidly in the global banking industry. A lot of financial institutions depend heavily on web-based applications to provide them with online banking services and these applications make it possible for users/customers to have access to their financial information, conveniently manage your account and to make transactions. Due to the increased dependence on these web technologies, banking applications/system are faced with cybersecurity threats. Examples of these cybersecurity threats are data breaches, phishing and unauthorised access to the system. These threats still continue to harm financial institutions across the global making security an essential part of modern software development. Companies can lower their system vulnerabilities and enhance the application's security through integrating security practices throughout the SDLC (Software Development Lifecycle)(OWASP, 2023).

In the past, security was frequently taken care of later in the development process especially during testing or deployment. Security weaknesses were often found too late as a result of this reactive strategy making their resolution more costly and difficult (Snyk, 2023). Modern software engineering methods suggest to include security measures from the planning, design, coding, testing and maintenance phases of the SDLC in order to handle this problem. This method is also known as DevSecOps or Secure SDLC which encourage to proactively detect and mitigate any security vulnerabilities during development (Snyk, 2023).

Customers can now access services via web and mobile applications because to banking industry's growing adoption of digital banking systems in South Africa. SABRIC (2024) states that although this development boosts efficiency and accessibility, it also makes people more vulnerable to cybercrime. digital banking fraud continues to be one of the most common types of financial crime with large financial losses reported yearly. Additionally, reports show that cybercrime related to banking has cost South Africa billions of rands emphasizing the critical need for more robust cybersecurity safeguards in digital financial systems (East Coast Radio, 2024).

Reference list

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